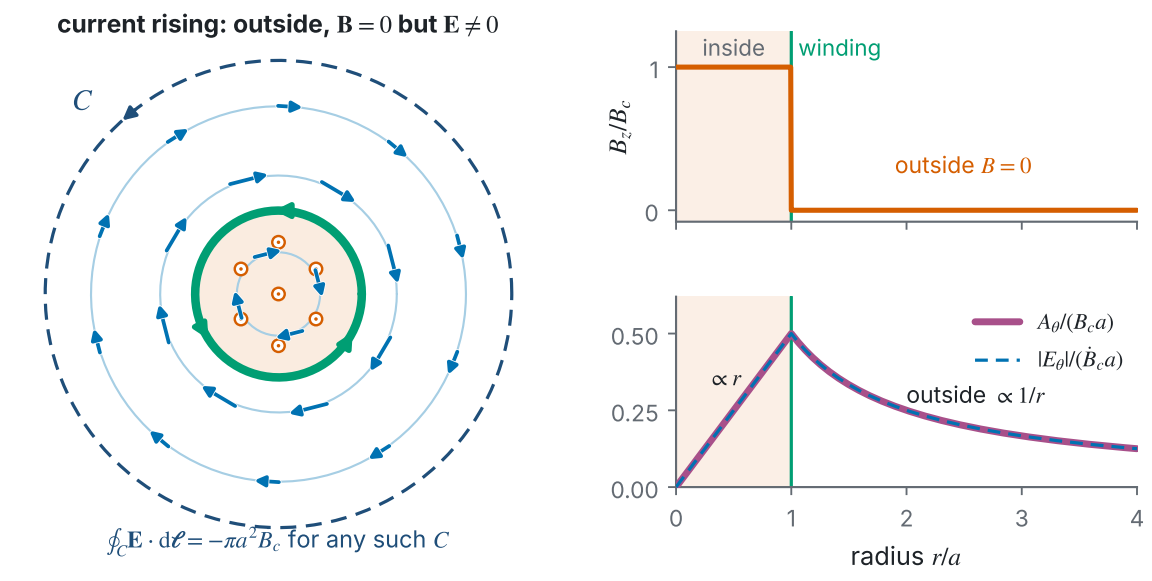
current rising: outside, $\mathbf{B} = 0$ but $\mathbf{E} \neq 0$
C
$\oint_C \mathbf{E} \cdot \mathrm{d}\boldsymbol{\ell} = -\pi a^2 \dot{B}_c$ for any such C
inside
winding
outside $B = 0$
B_z/B_c
1
0
$A_\theta/(B_c a)$
$|E_\theta|/(\dot{B}_c a)$
$\propto r$
outside $\propto 1/r$
0.50
0.25
0.00
0
1
2
3
4
radius r/a